

Toka Alokda, M.Sc.

Ph.D. researcher in Cosmology

Argelander-Institut für Astronomie, University of Bonn

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Links [LinkedIn](#) - [Github](#) - [Website](#)

Expertise and research interests

- **Cosmology:** primordial non-Gaussianity; Large-Scale structure: formation, summary statistics, perturbation theory, numerical simulations.
 - **Astrostatistics:** Bayesian inference; Reliability of cosmological data analysis and inference pipelines.
 - **Machine learning:** deep neural networks for scientific data analysis; neural simulation-based inference (SBI); interpretable AI for scientific discovery.
 - **High-performance computing:** GPU-accelerated code development; differentiable programming; scalable scientific software and applications.
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Work experience

DFG Cluster of Excellence “Our Dynamic Universe” (DYNAVERSE)

PhD researcher. **Apr. ‘26 - Present**

Argelander-Institut für Astronomie, University of Bonn

PhD researcher. **Jun. ‘24 - Present**

University of Bonn - Astrophysics MSc program

Graduate teaching assistant. **Oct. ‘22 - Feb. ‘24**

Education

University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn) - Bonn, DE.

Ph.D. - Astrophysics.

International Max Planck Research School for Astronomy and Astrophysics (IMPRS A&A).

Topic: Exploring Primordial Non-Gaussianity through the Large Scale Structure with Machine Learning.

Advisor(s): Prof. Dr. Cristiano Porciani, Dr. Alexander Eggemeier.

June ‘24 - Present

University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn) - Bonn, DE.

M.Sc. - Astrophysics.

Bonn-Cologne Graduate School for Physics and Astronomy.

Thesis: “Identification of Protohalos with Deep Learning” (Grade: 1.0¹)

Grade: 1.6 (ranking: 3rd in the class of 2023).

Oct. ‘21 - Dec. ‘23

¹ Indicated in the German system; whereas 1.0 is the highest grade and 5.0 is the lowest.

University of Science and Technology at Zewail City - Giza, EG.

B.Sc. - Physics of Earth and Universe; Astrophysics concentration.

GPA: 3.63 ² (Cum Laude - final semester: 4.0/4.0)

Thesis: “A Dark Matter Explanation for Galactic Positron Flux” (Grade: A)

Sep. ‘17 – Jul. ‘21

Awards & Honors

Bonn-Cologne Graduate School of Physics and Astronomy (BCGS) Scholarship

Oct. ‘21 - Sep. ‘23

B.Sc. Honors

- **Cum Laude** - Awarded for being in the top 10% of the graduating class.
- **Provost’s Honors Roll** - Awarded on a semesterly basis to the top 10%.

Jul. ‘21

Zewail City UST Scholarship

Full scholarship awarded to the top students in the UST admissions exams.

Sep. ‘17 - Jul. ‘21

Egyptian national secondary certificate “Thanaweya Amma” distinction

Awarded to the top 0.0% in the Egyptian national “Thanaweya Amma” exams.

Jul. ‘17

Conference presentations

“Coverage is not enough: What SBI posteriors of f_{NL} are actually telling you”

Waterloo, CA - Contributed talk in the “Advancing simulation-based and field-level inference for Cosmology” conference.

Jun. ‘26

“The quest for primordial non-Gaussianity: Opportunities and Challenges”

Sofia, BG - Contributed lightning talk at the CosmoVerse 2026 workshop.

May ‘26

“Segmenting Protohalos with Vision Transformers”

Cologne, DE - Contributed talk in the Astroinformatics 2026 conference.

Feb. ‘26

“Identification of Protohalos with Deep Learning”

Les Houches, FR - Contributed talk at the Les Houches - Dark Universe school.

Jun. ‘25

“Identification of Protohalos with Deep Learning”

Cambridge, MA - Online poster presentation in the first AstroAI workshop.

Jun. ‘24

Seminar and Colloquium presentations

“Identification of Protohalos with Deep Learning”

Bonn, DE - Poster presentation in the annual BCGS poster session.

Apr. ‘24

“CNNs and ViTs for Cosmological Structure Formation”

Bonn, DE - Invited talk in the “Machine Learning Seminar”.

Dec. ‘23

“Identification of Protohalos with Deep Learning”

Bonn, DE - Master’s colloquium in the Lens and Cosmology seminar at AIfA, Bonn.

Nov. ‘23

² Indicated in the American system; whereas 4.0 is the highest grade and 0.0 is the lowest.

Publications

Alokda, T., Porciani, C. and Eggemeier, A. (2026). *Coverage is not enough: Frequentist tests of simulation-based inference for primordial non-Gaussianity.* arXiv: 2605.00980

Alokda, T. and Porciani, C. (2025). *Segmenting proto-halos with vision transformers.* JCAP 11 (2025) 083.

Technical skills

- **Programming & Scientific Computing:**
Python (advanced), C++ (advanced), Julia (intermediate), MATLAB (intermediate), Wolfram Mathematica (intermediate), Fortran (basic).
 - **Machine Learning & Differentiable Programming:**
PyTorch, TensorFlow, JAX, Keras, Optuna, Kymatio, SciPy.
 - **Astrophysics & Cosmology Tools:**
GADGET-4, L-Galaxies, AHF, CLASS, PINOCCHIO, Python libraries for cosmological analysis: e.g. Pylans, PySpectrum, Nbodykit.
 - **High-Performance Computing & OS:** Linux, Bash/UNIX shell, Slurm
 - **Documentation & Reporting:** LaTeX, MS Office
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Teaching Experience

University of Bonn

Oct. '22 - Present

Graduate Teaching Assistant

Summer Semester 26

The large scale structure as revealed by redshift surveys (M.Sc. of Astrophysics)

Winter Semester 22/23, 23/24, 24/25, and 25/26

Maths for Theoretical Astrophysics and Cosmology (M.Sc. of Astrophysics).

Summer Semester 24 and 25

Advanced Topics in Cosmology (M.Sc. of Astrophysics).

Summer Semester 23

Astrophysics of Galaxies (M.Sc. of Astrophysics).

University of Science and Technology at Zewail City

Sep. '20 - Jan. '21

Undergraduate Teaching Assistant

Fall 2021: Quantum Mechanics I (B.Sc. of Physics).

Public Outreach

“Tangled in the Web: The rise of AI in Astronomy” - Astronomy on Tap - Bonn, DE

Oct. '24
